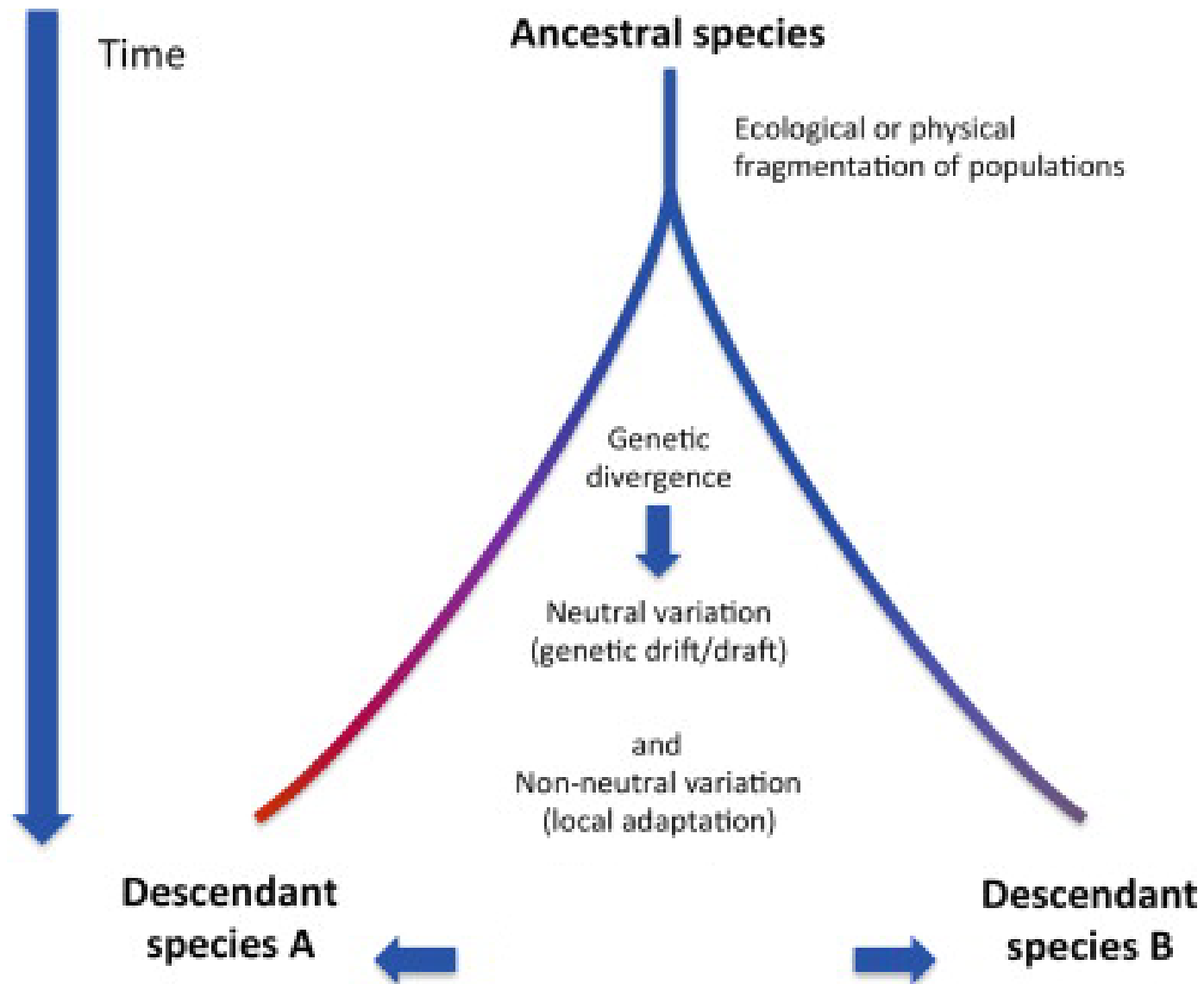
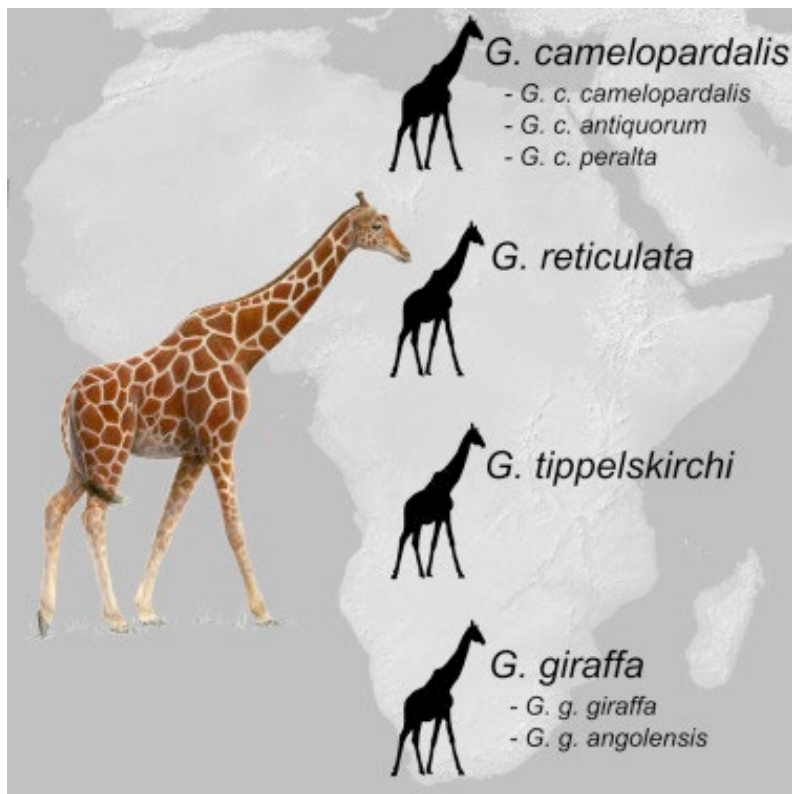


Selection and selection scans

Mpala Workshop, August 2026





We often want to know whether selection has acted to differentiate the populations.
In other words, do populations have unique adaptations that we should know about.



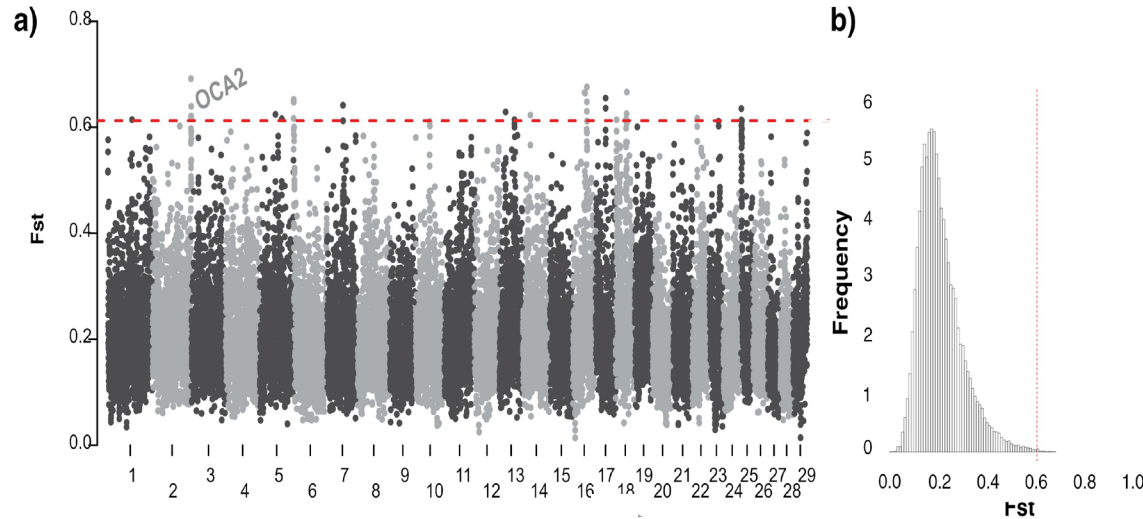
Why selection is relevant for conservation

- Identifying populations with unique adaptations.
- Managing population connectivity and genetic rescue.
- Predicting climate change resilience.
- Predicting disease resistance and vulnerability.

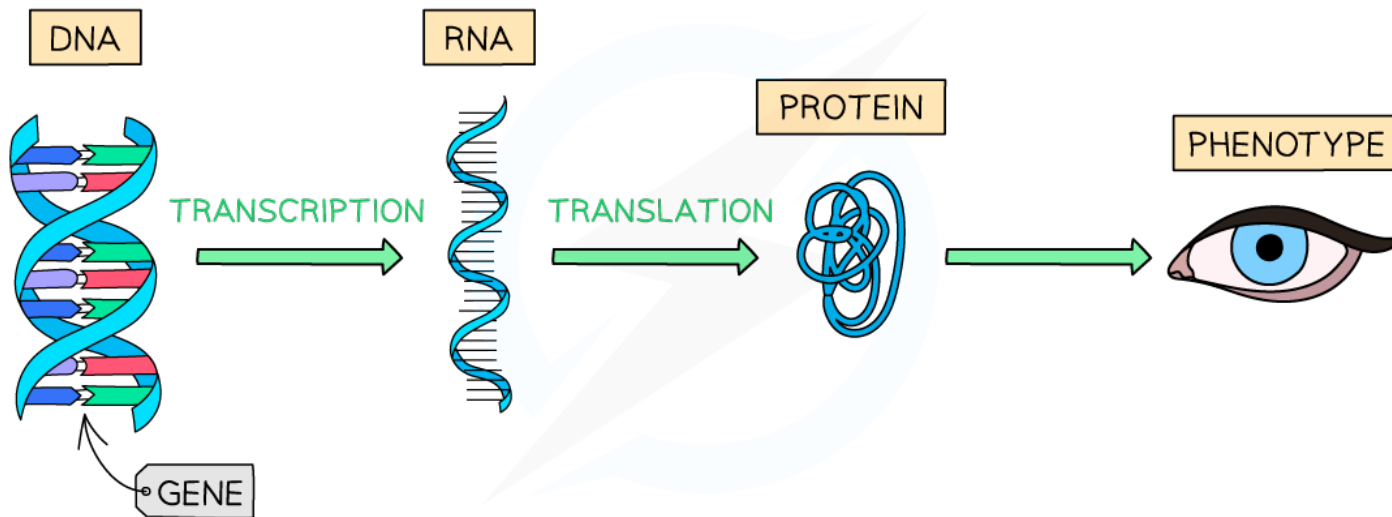
Several of the following slides were stolen by me from Anders Albrechtsen, Matteo Fumagalli and Cindy Santander



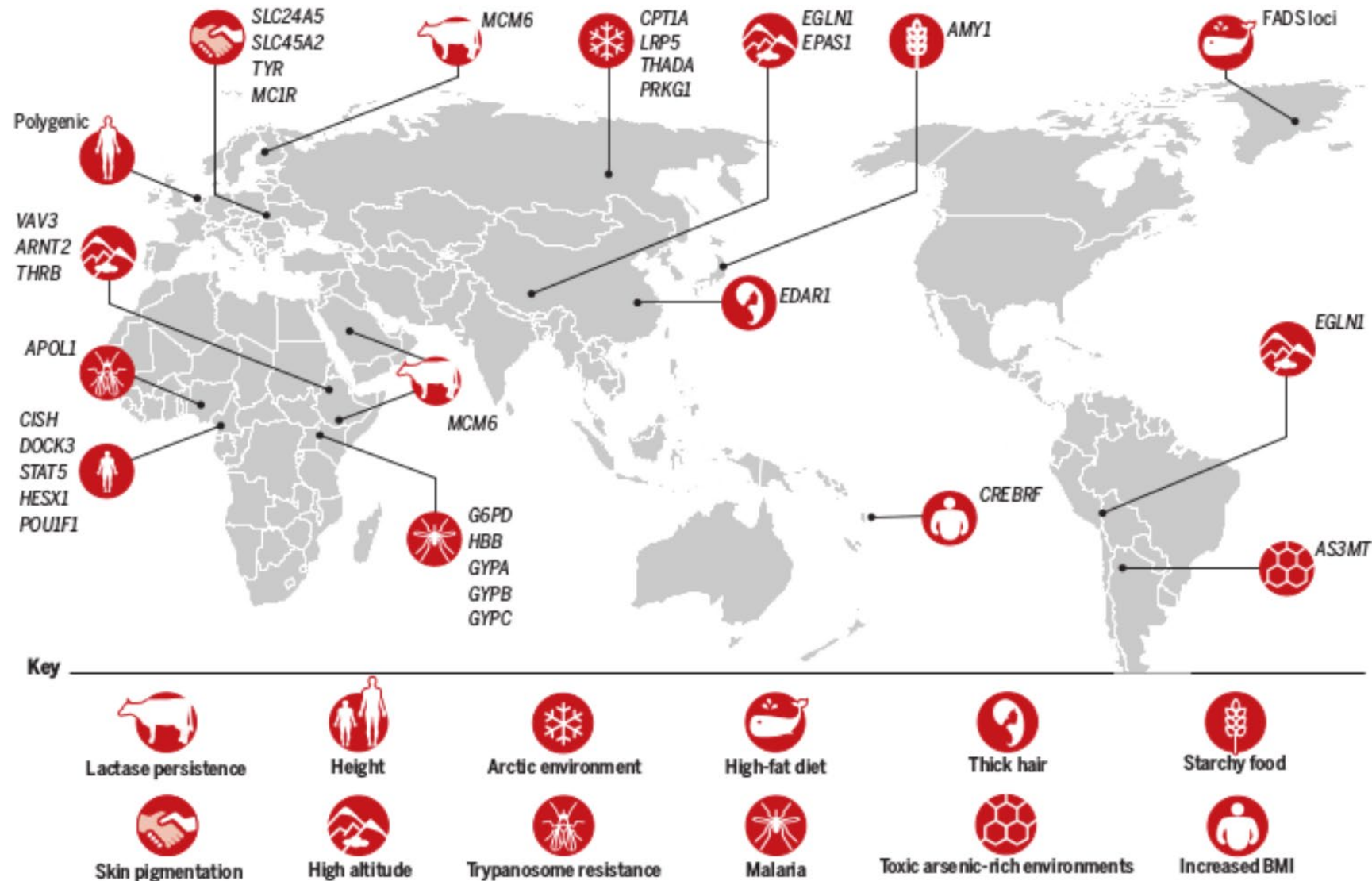
Selection is not easy to identify



1. Detect genomic outliers.
2. Assess significance (reject other explanations than selection).
3. Ideally: understand function and benefit.

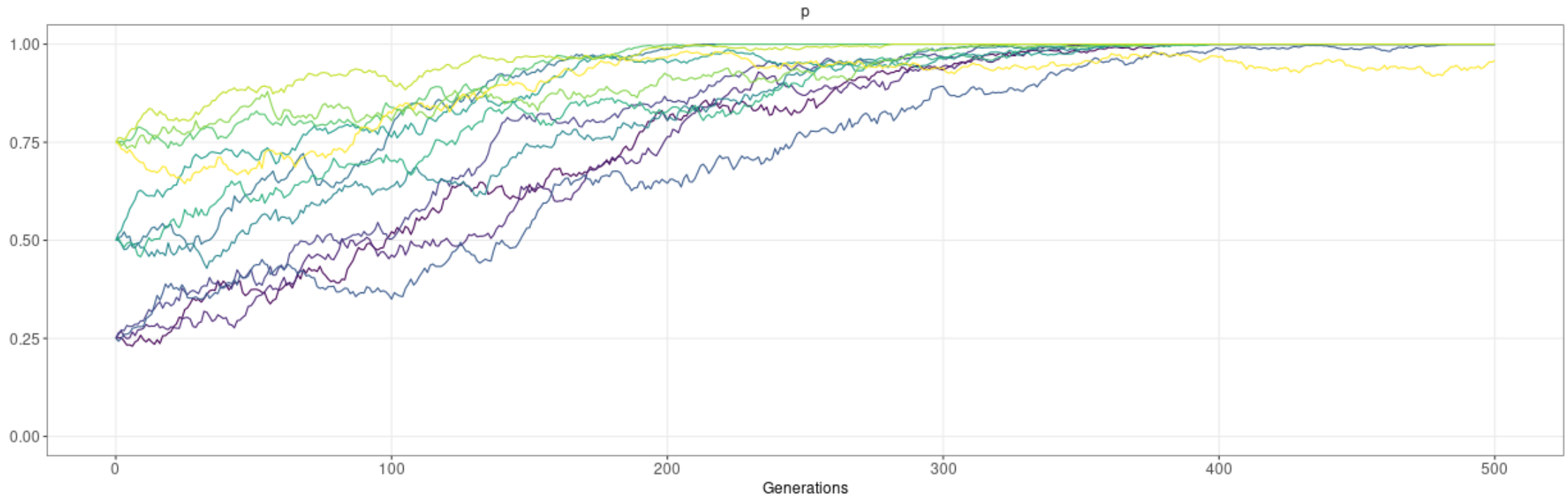


Examples of genes under selection in humans



Positive selection

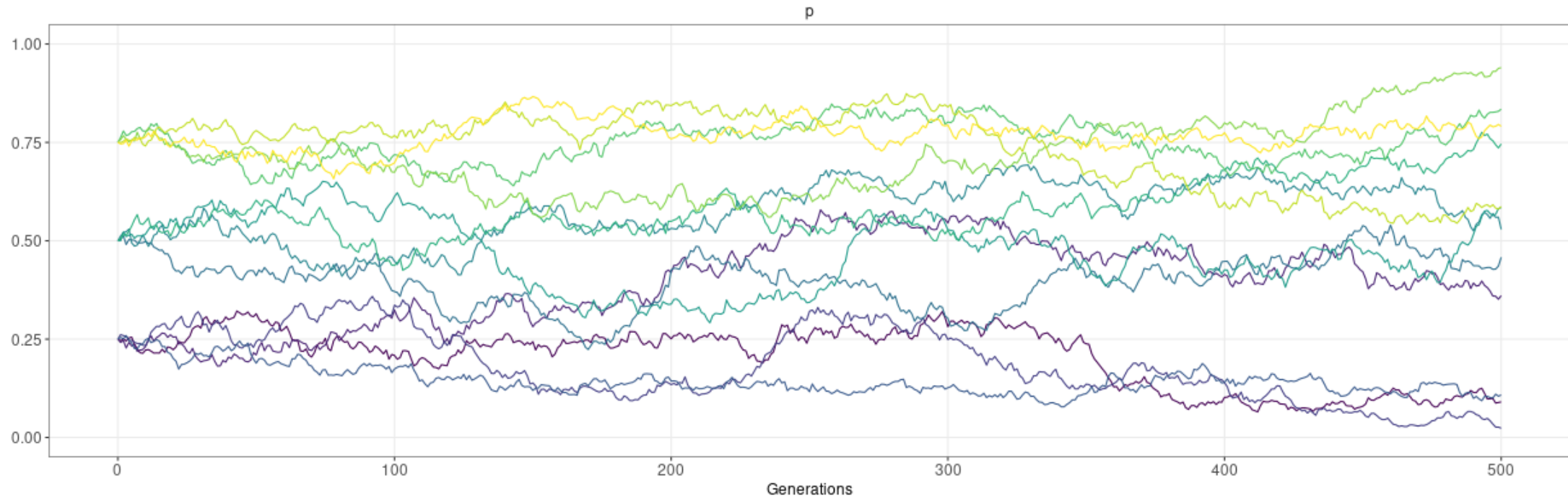
$N=1000$, $s=0.01$ (high!!)



A beneficial allele increases in frequency

Neutral evolution

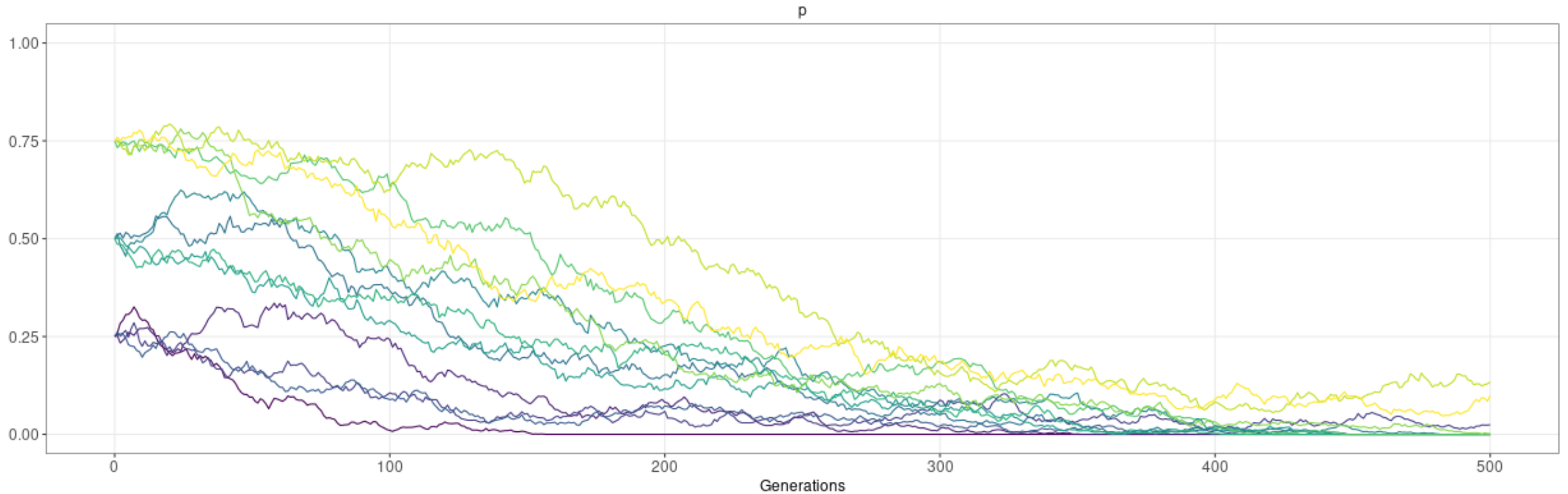
$N=1000$, $s=0.00$



A neutral allele decreases OR increases in frequency

Negative selection

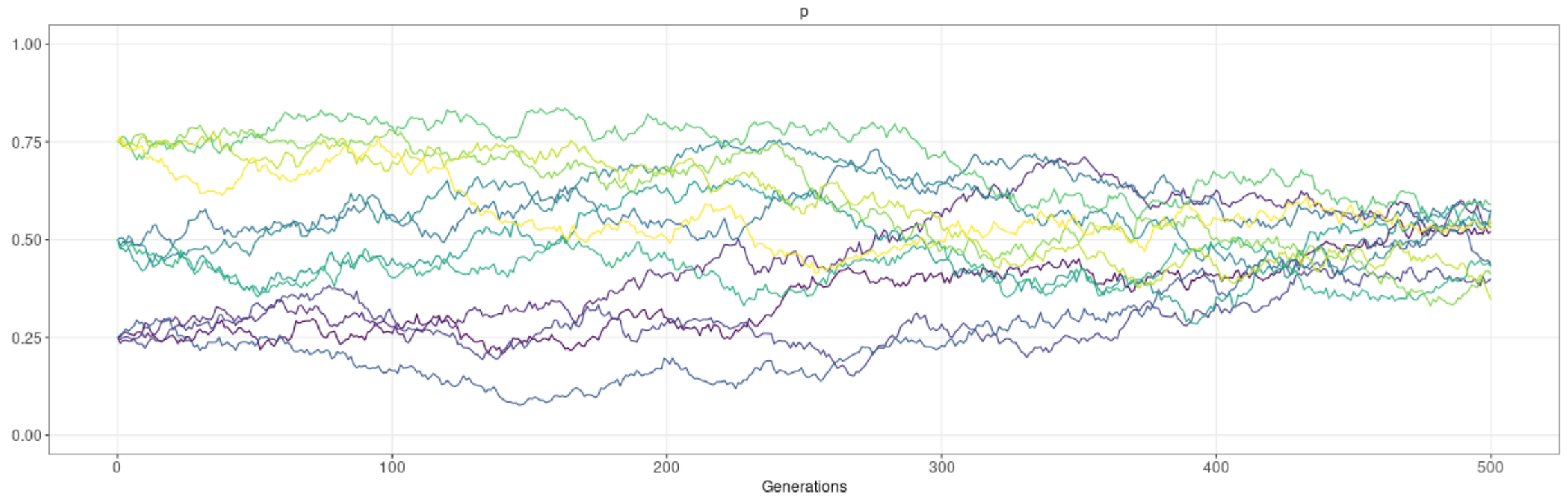
$N=1000$, $s=-0.01$



A harmful allele decreases in frequency

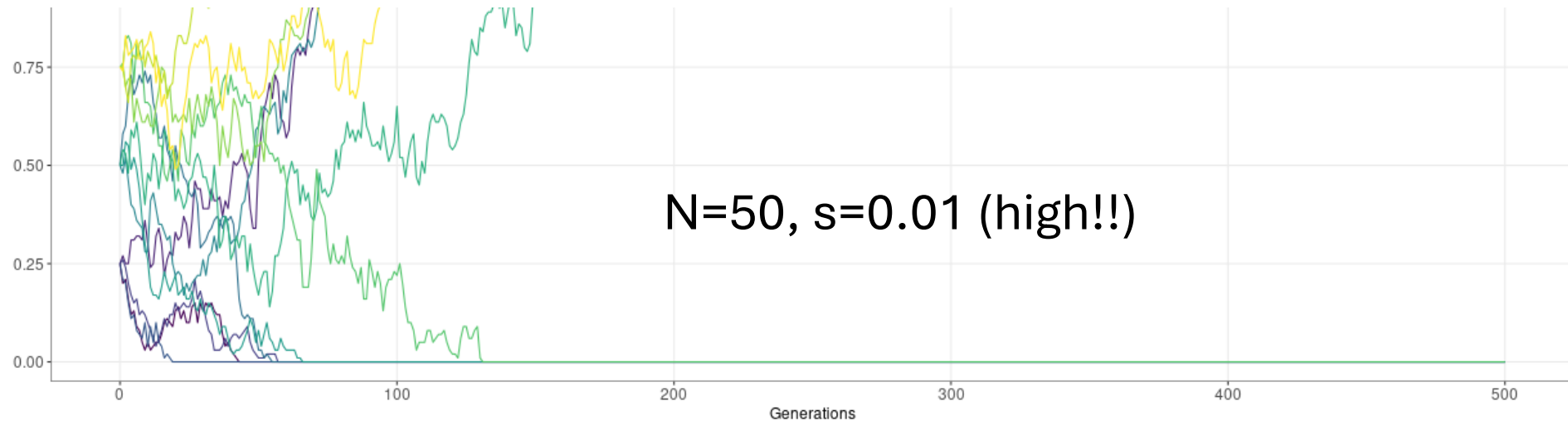
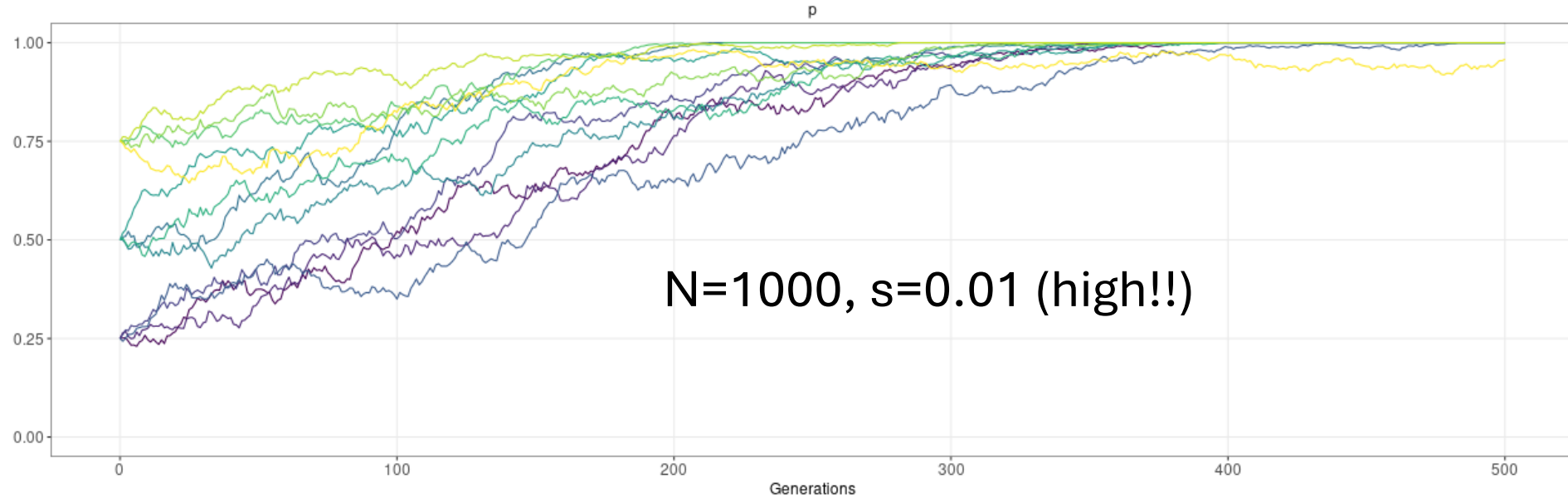
Balancing selection

$N=1000$, $s=-0.05$ and $t=-0.05$



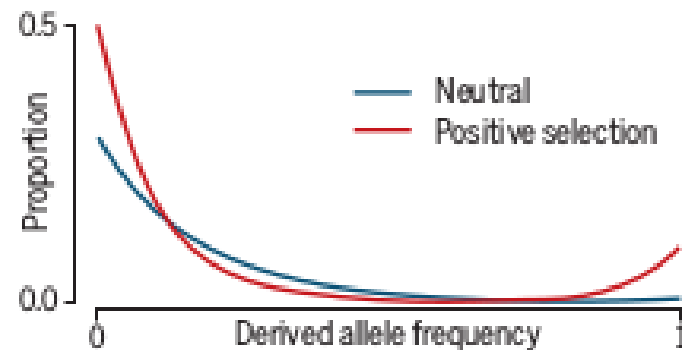
Heterozygotes are favored, both alleles are maintained

But remember pop size also matters!

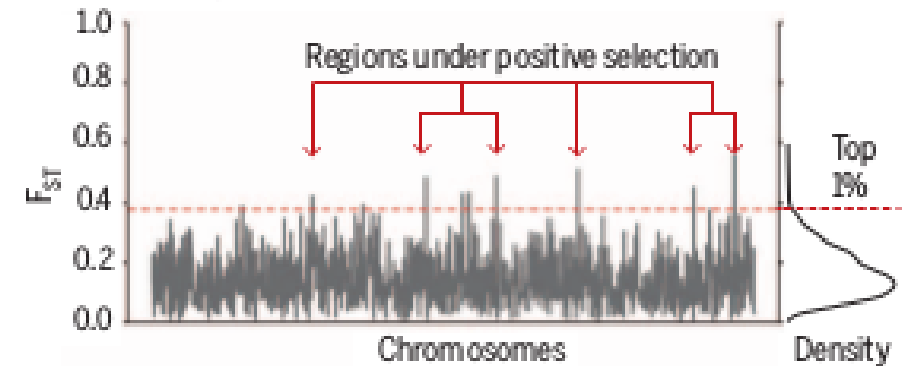


Methods for detecting selection

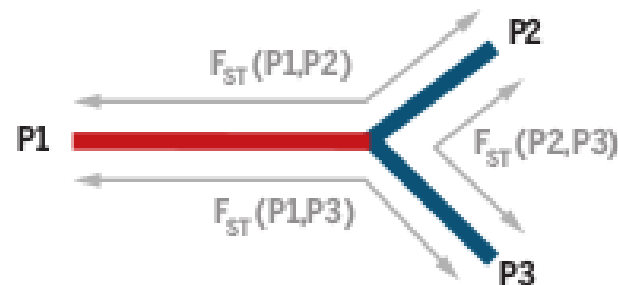
i) Change in allele frequency spectrum



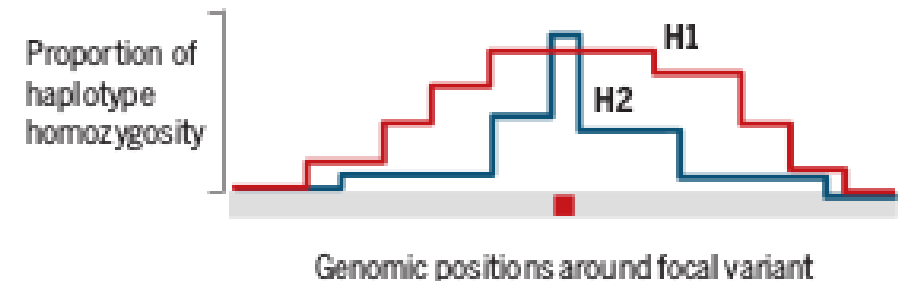
ii) Change in F_{ST} along genome



iii) Locus-specific branch length (LSBL)



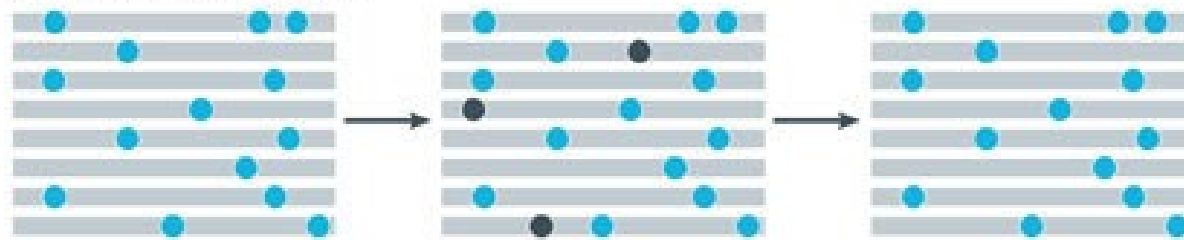
iv) Extended haplotype homozygosity (EHH)



Past

Present

a Purifying selection

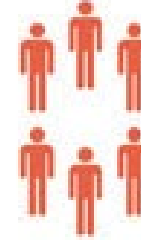
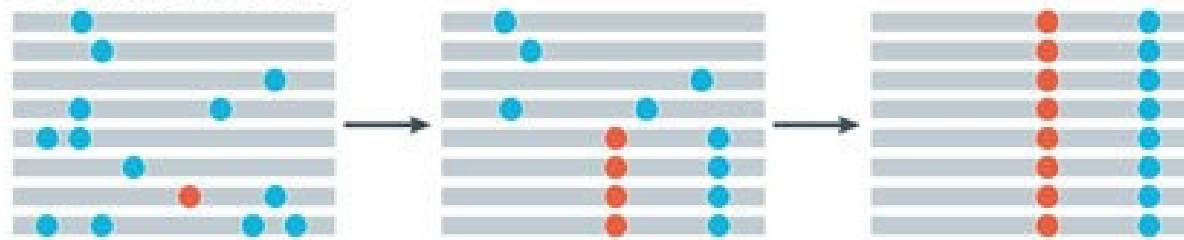


Advantaged population

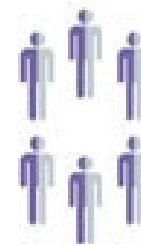
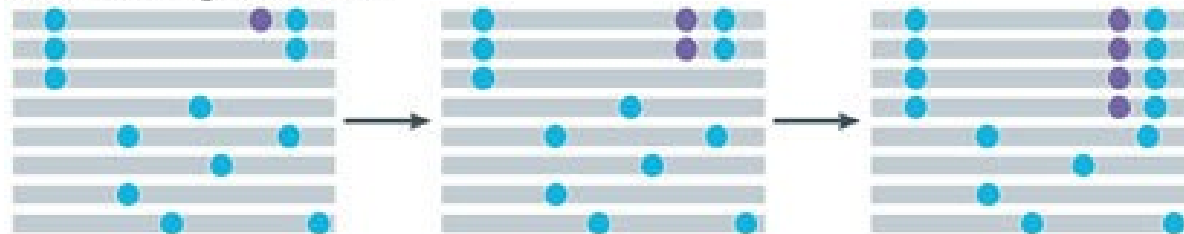
Disadvantaged individual or population



b Positive selection

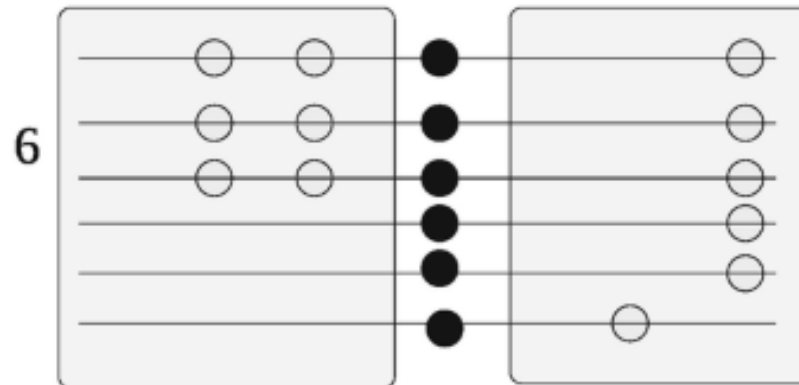
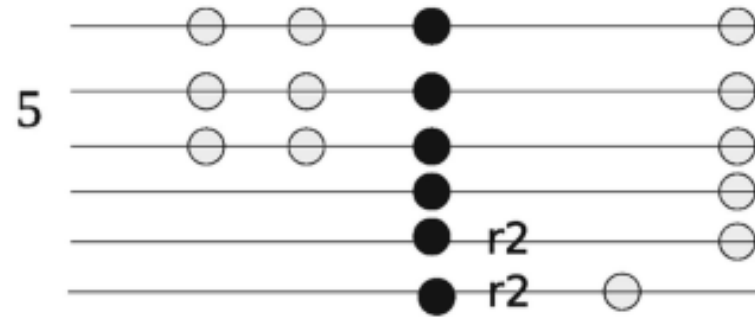
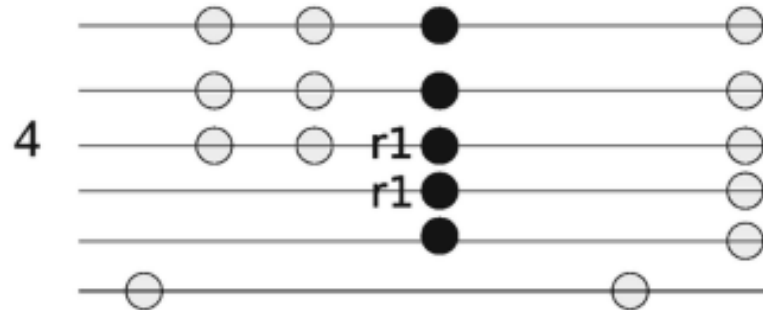
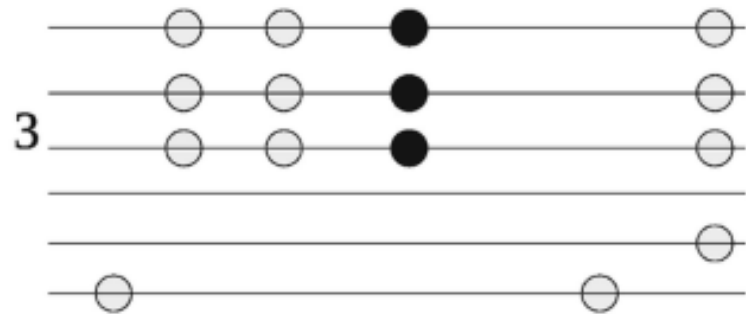
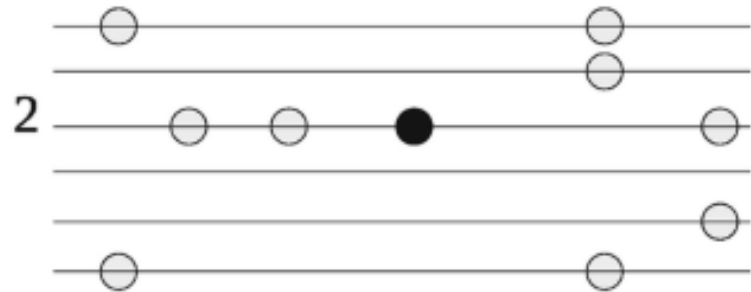
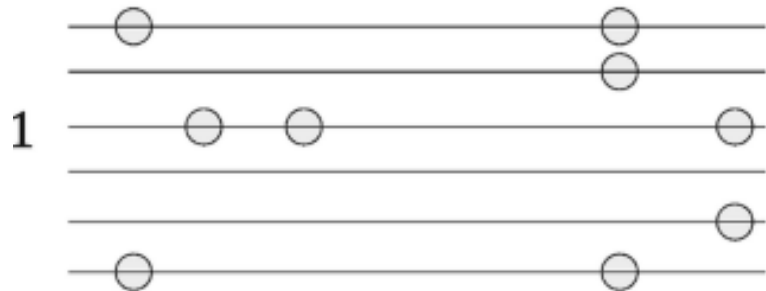


c Balancing selection

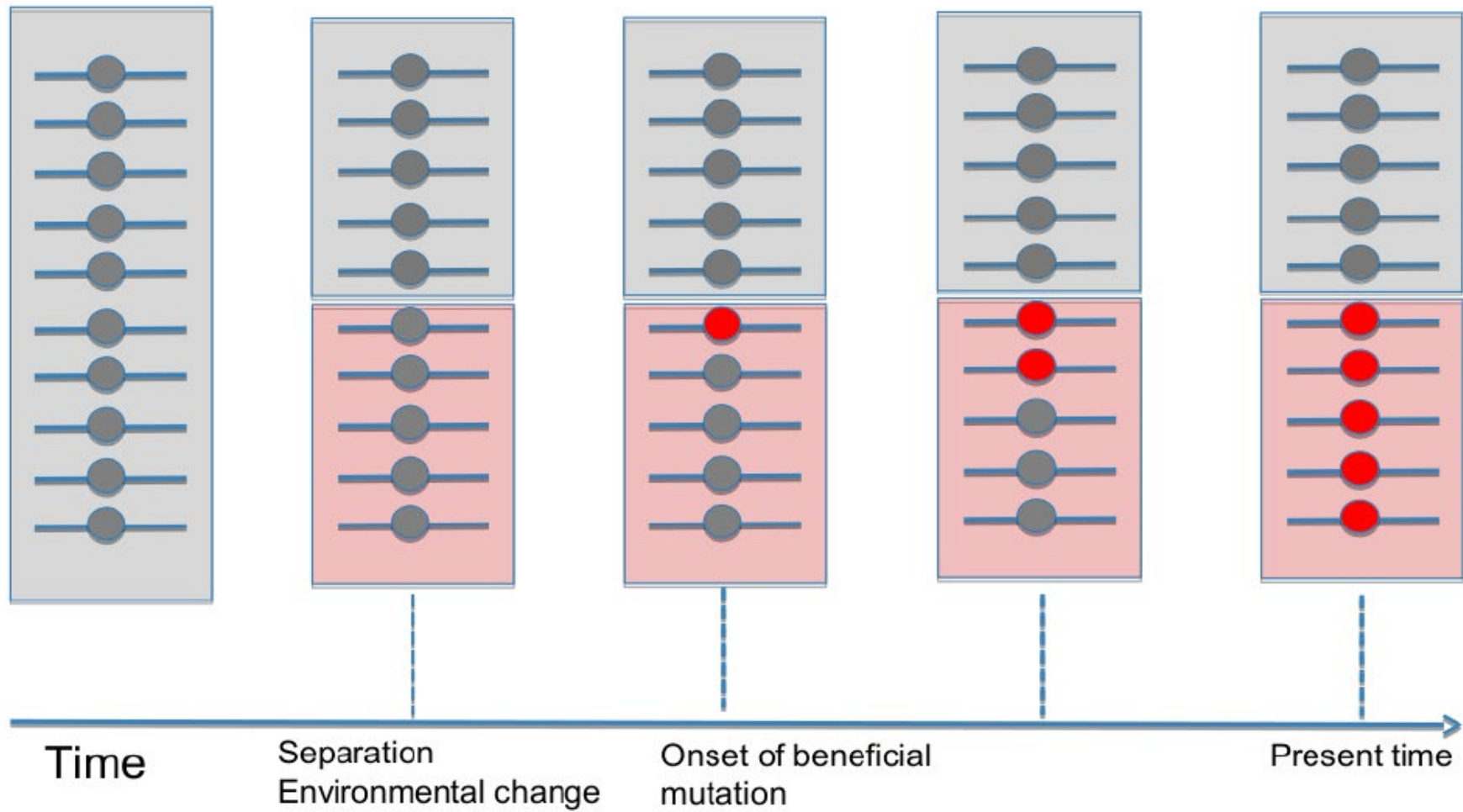


- Neutral mutation
- Positively selected mutation
- Deleterious mutation
- Balanced mutation

Selective sweep

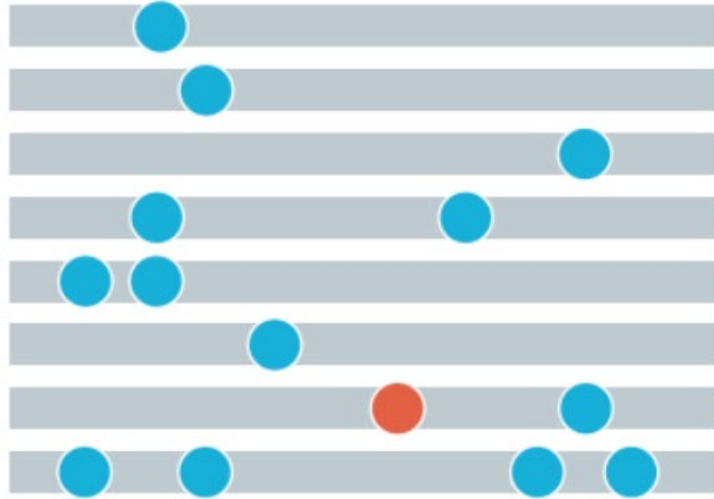


r1 and r2 mark two independent recombination events



How will F_{st} between these two populations change?
Will the F_{st} change affect the whole genome?

Before selection



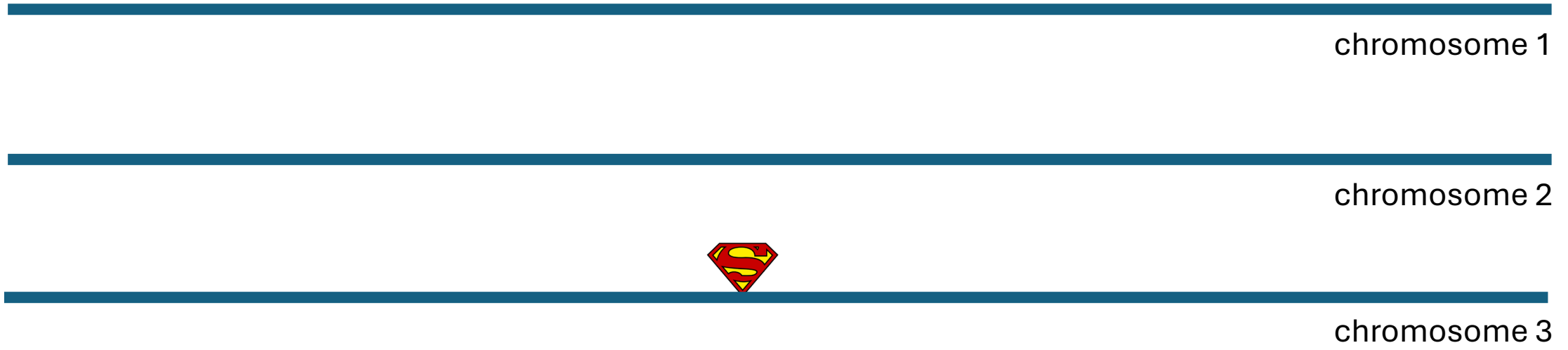
↑
ADVANTAGEOUS
MUTATION

After selection

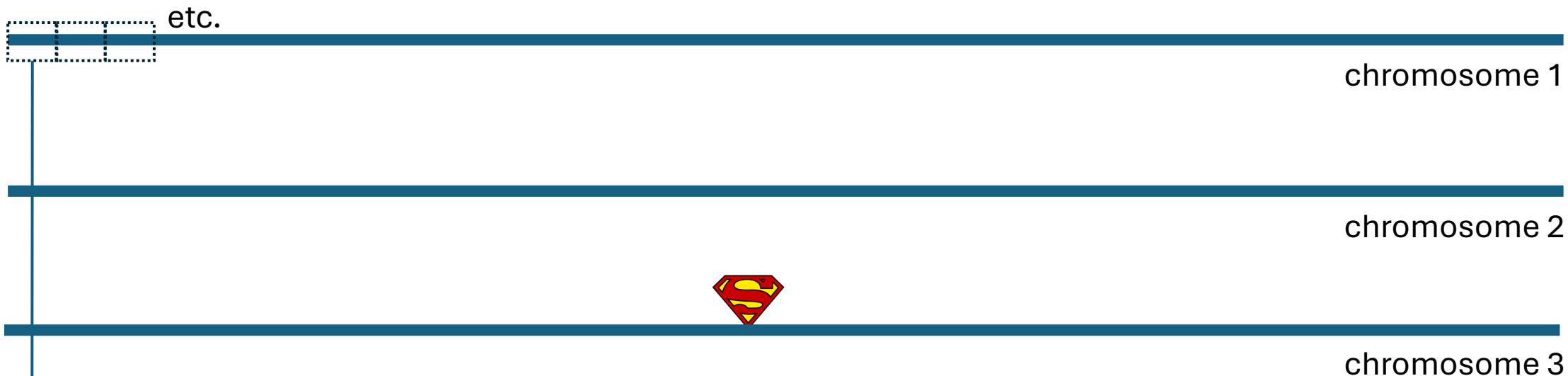


↑
ADVANTAGEOUS
MUTATION

How would we detect a genomic region
where this has occurred?

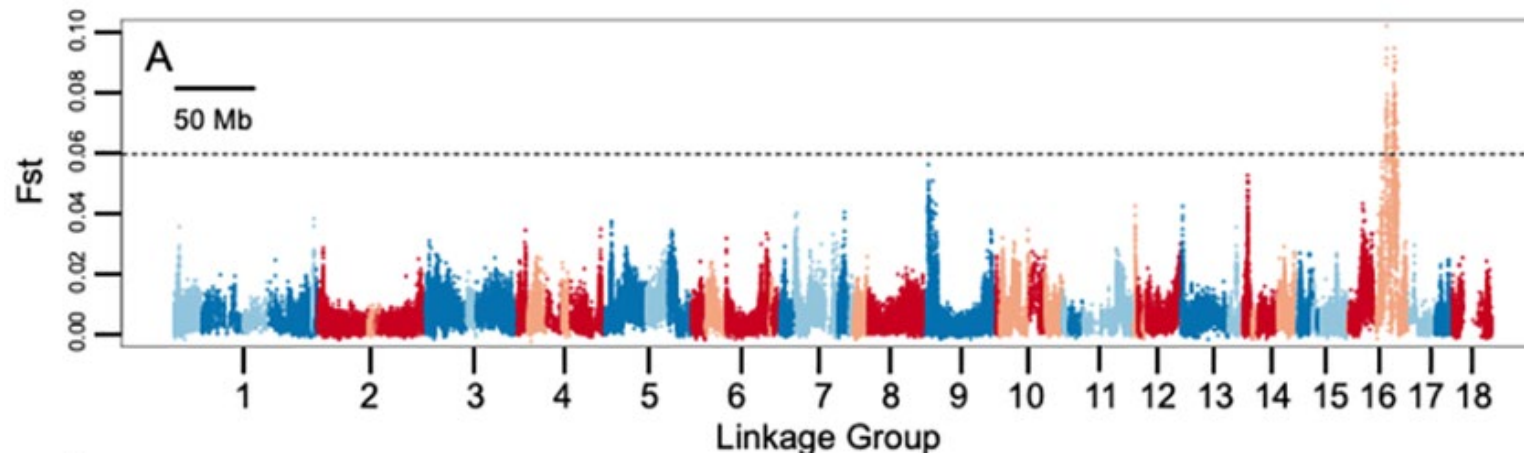


How do we find the superman gene?

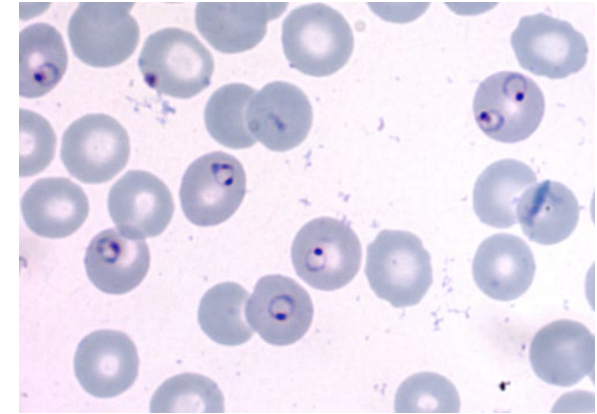
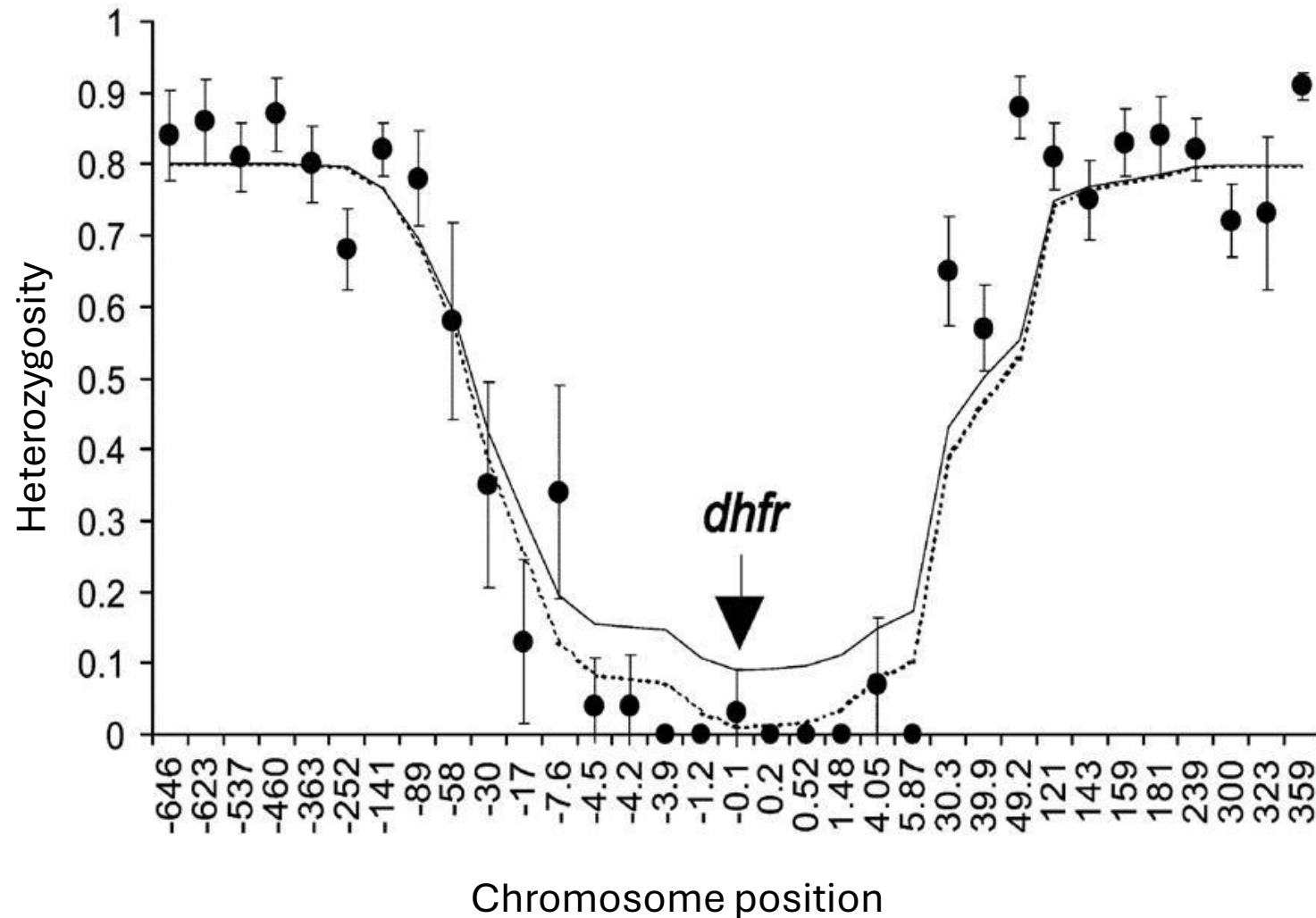


We can move along the genome and calculate some relevant statistics

- P_i
- F_{ST}
- PBS
- SFS
- Haplotype stats



Genetic diversity decreases locally

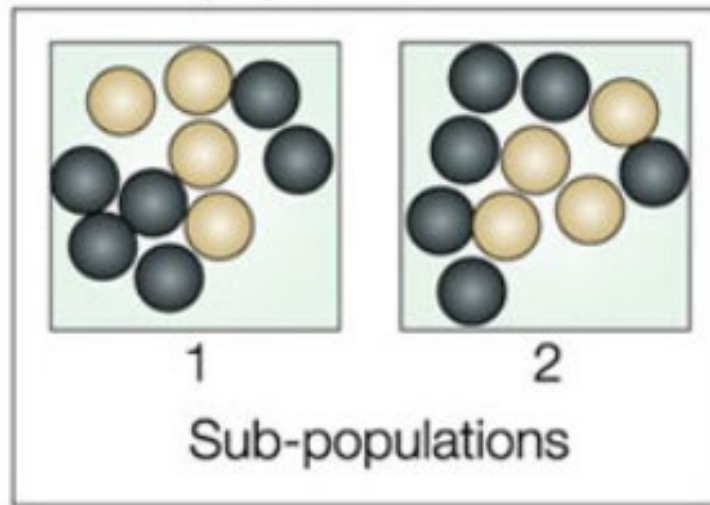


A mutation in the gene *dhfr* confers resistance to fansidar, an anti-malaria medication

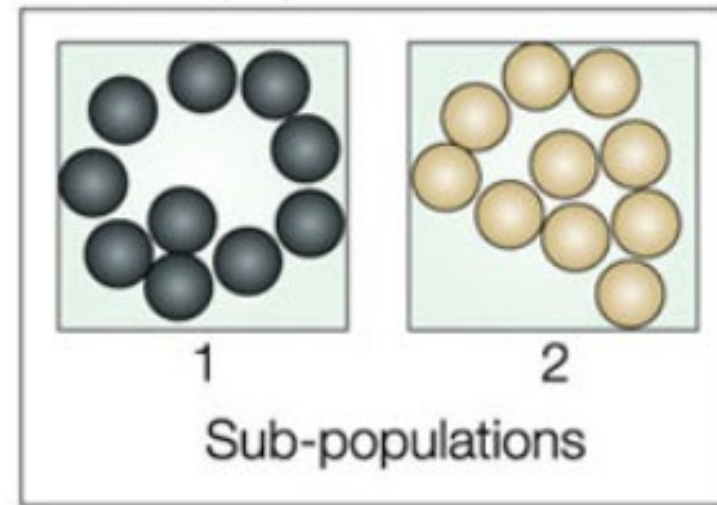
F_{ST} is often used for selection scans

Fst from heterozygosity

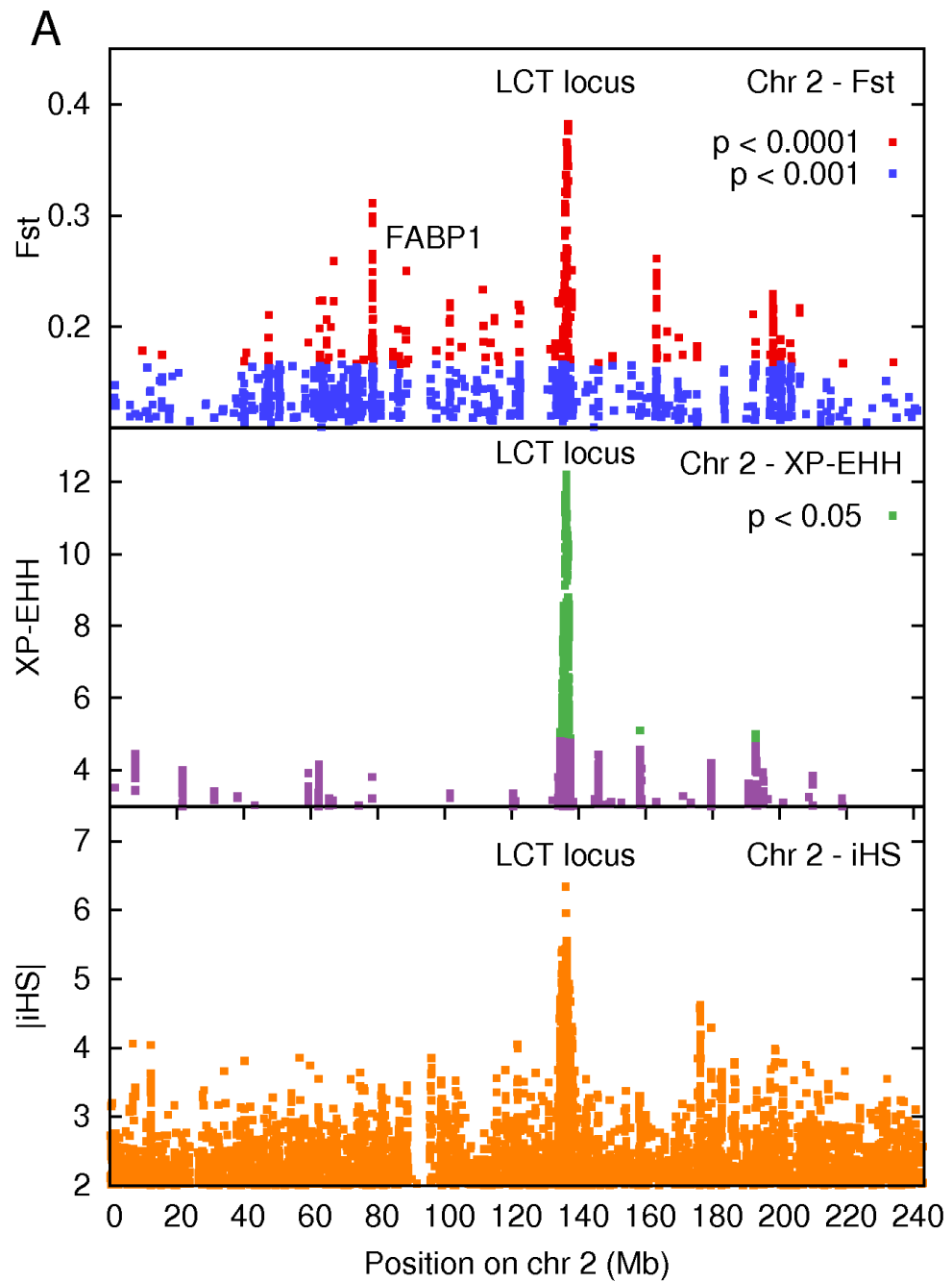
$$F_{st} = \frac{\sigma_B}{\sigma_T} = \frac{H_{total} - H_{subpopulations}}{H_{total}}$$

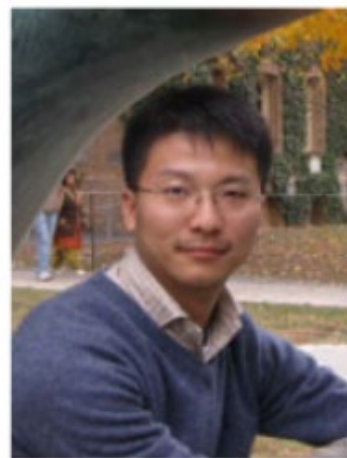


$F_{ST} = 0$



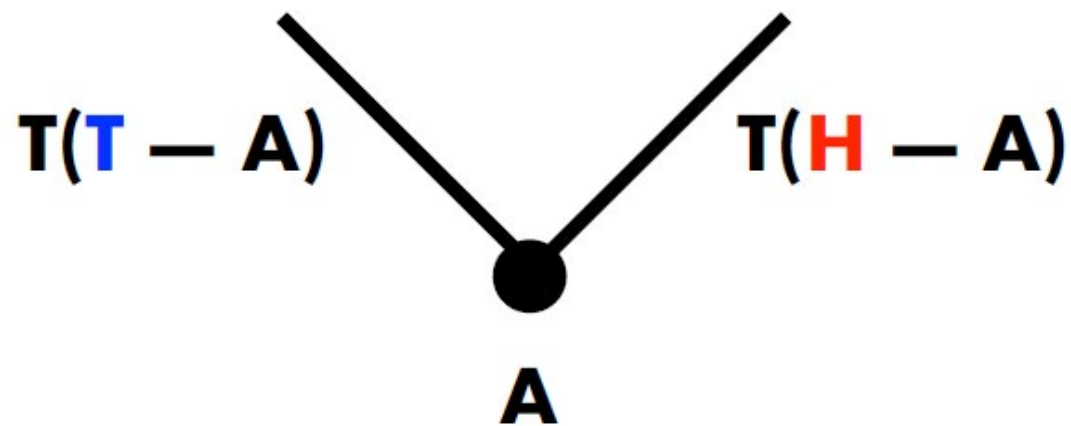
$F_{ST} = 1$





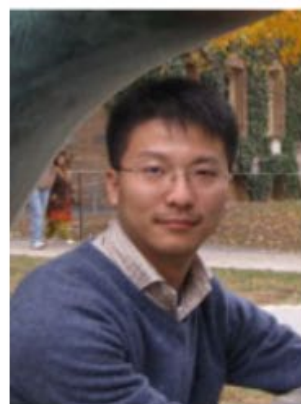
T

H

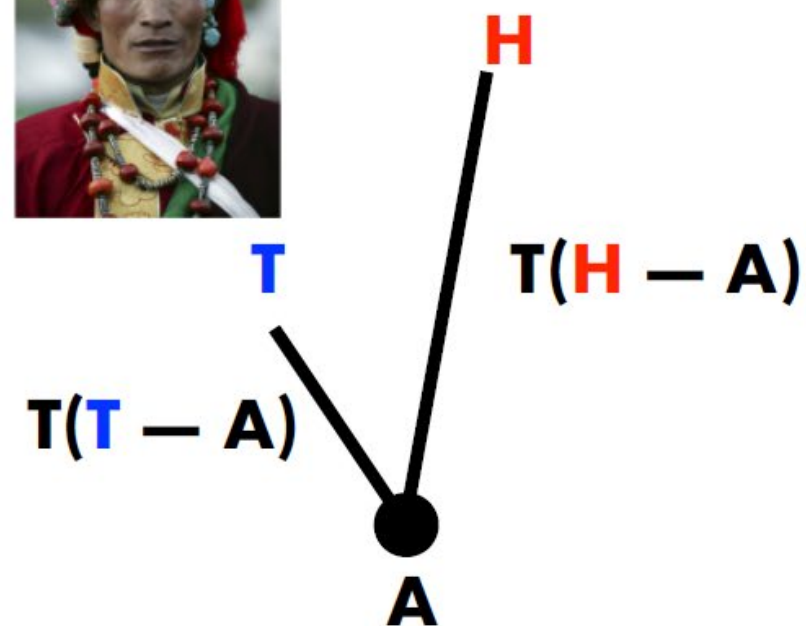
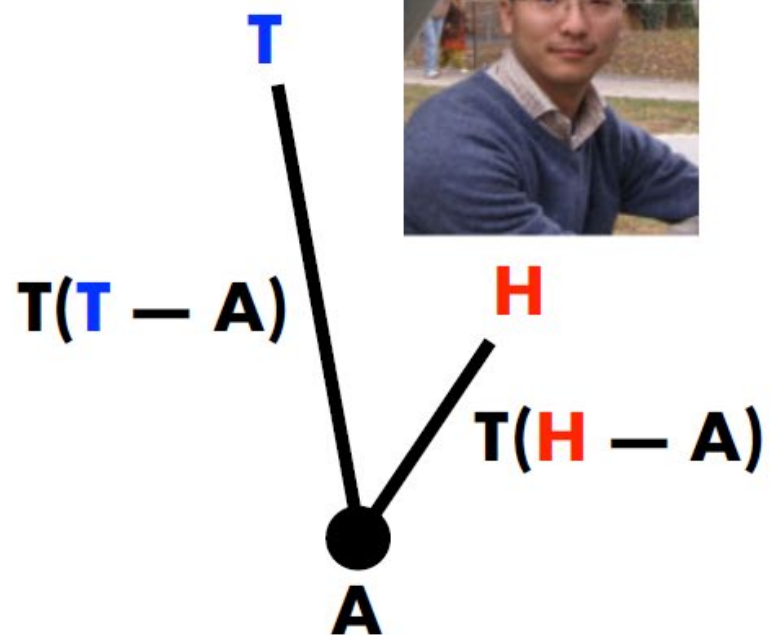
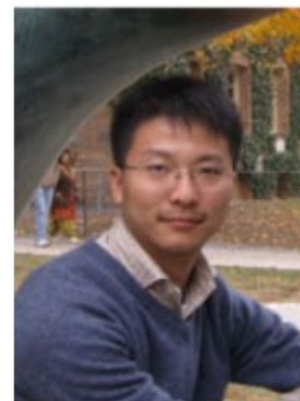


$$F_{ST}(\mathbf{T} - \mathbf{H}) \sim \mathbf{T}(\mathbf{T} - \mathbf{A} - \mathbf{H})$$

$$F_{ST}(\mathbf{T} - \mathbf{H}) \sim \mathbf{T}(\mathbf{T} - \mathbf{A} - \mathbf{H})$$

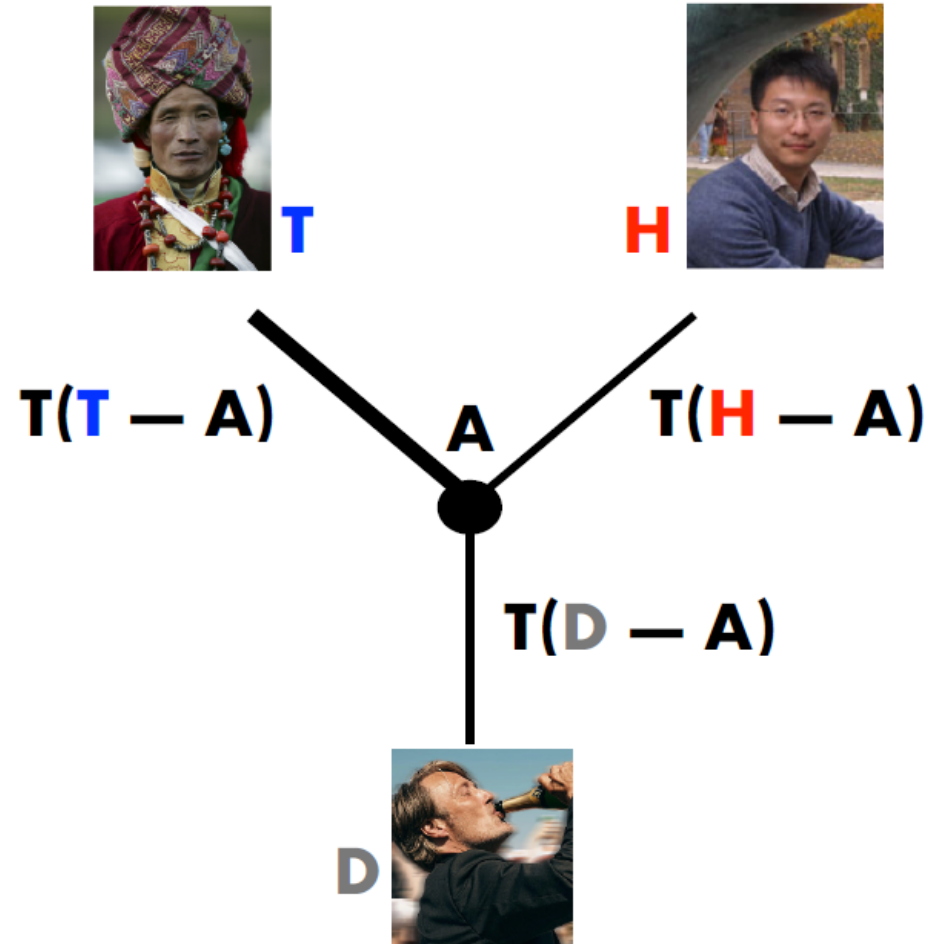


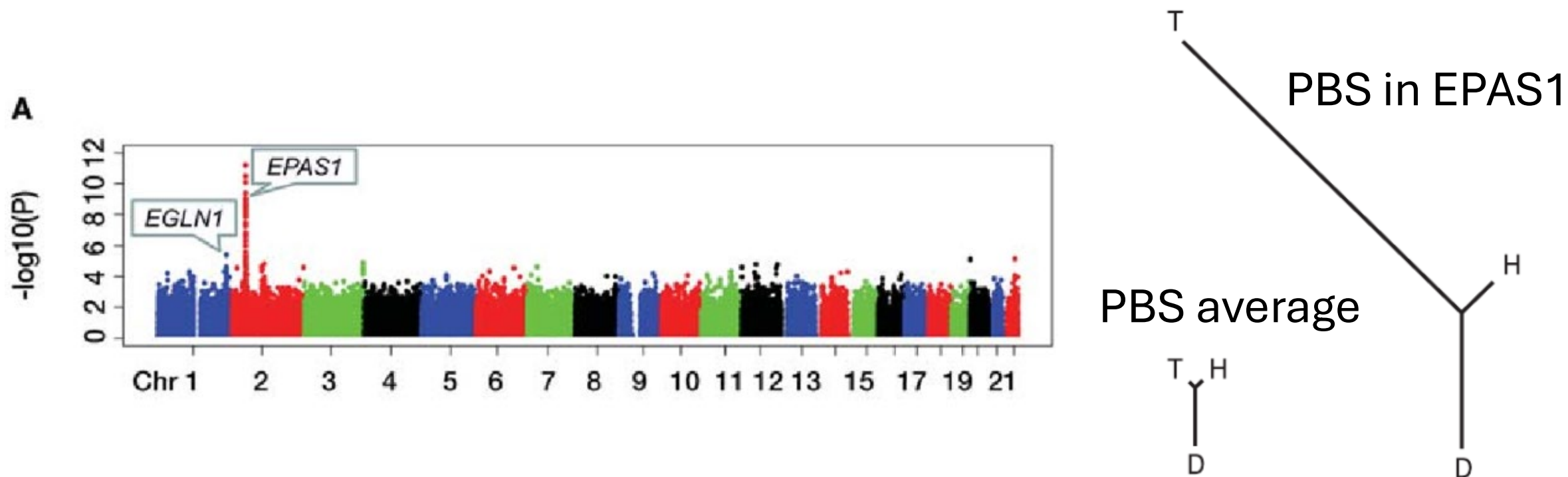
?



Introducing the Population Branch Statistic

$$PBS = TBS = (T^{TH} + T^{TD} - T^{HD})/2, \quad T^{AB} = -\log(1 - F_{st}^{AB})$$



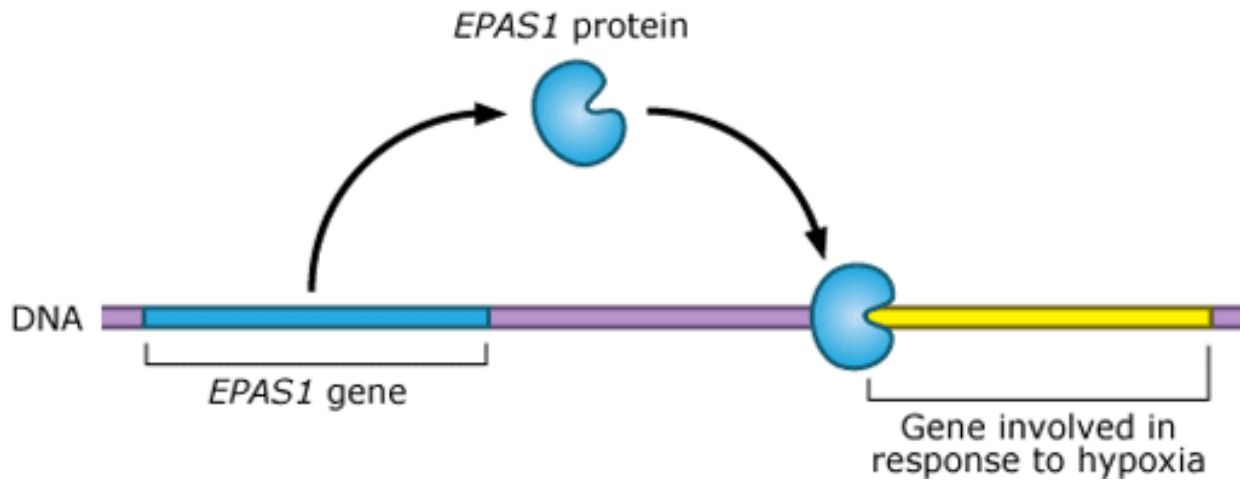


PBS for Tibetans is highly elevated in the
EPAS1 gene

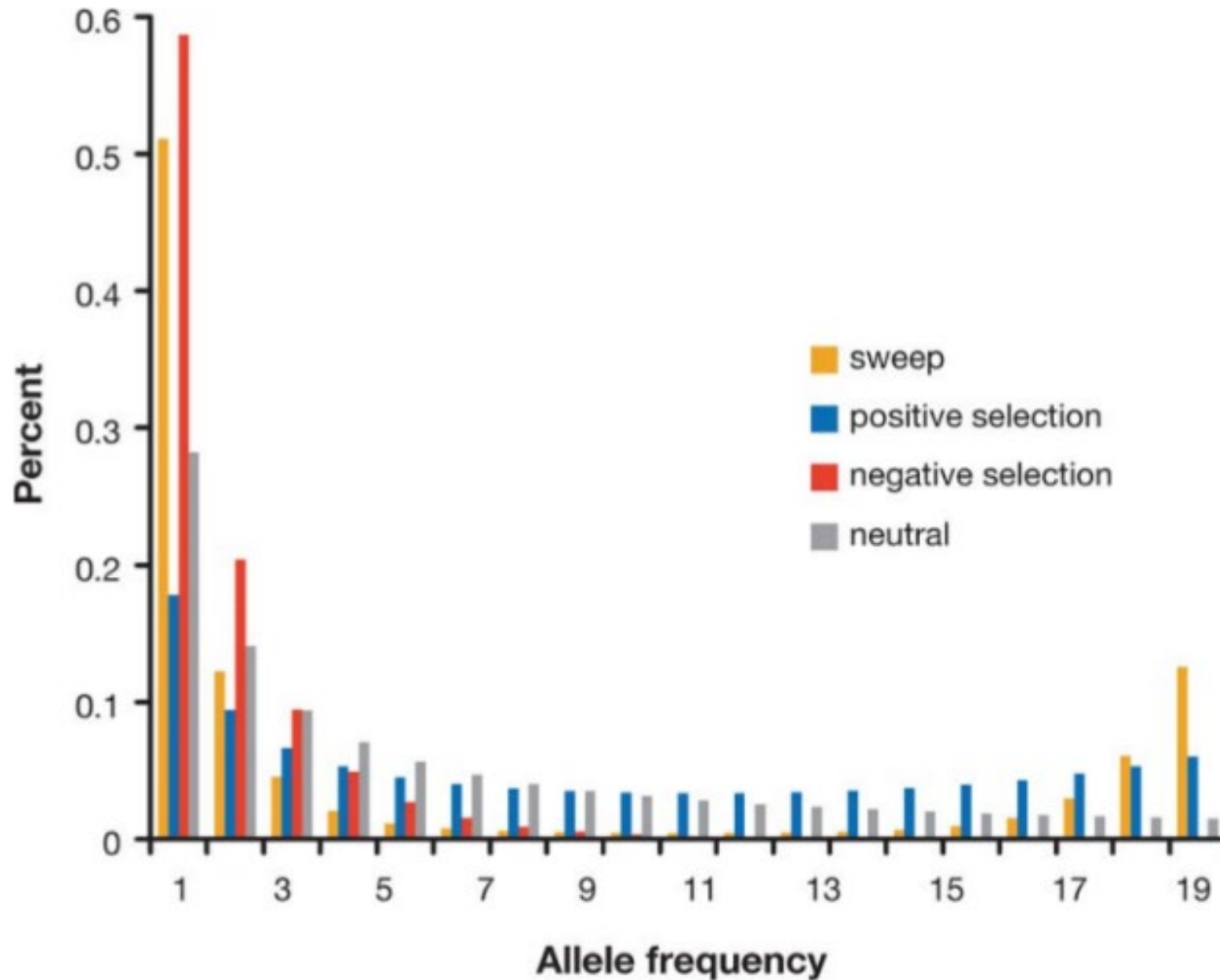
EPAS1 SNP allele frequencies

Allele	Tibetan	Han	Danish
C	0.13	0.9125	1
G	0.87	0.0875	0

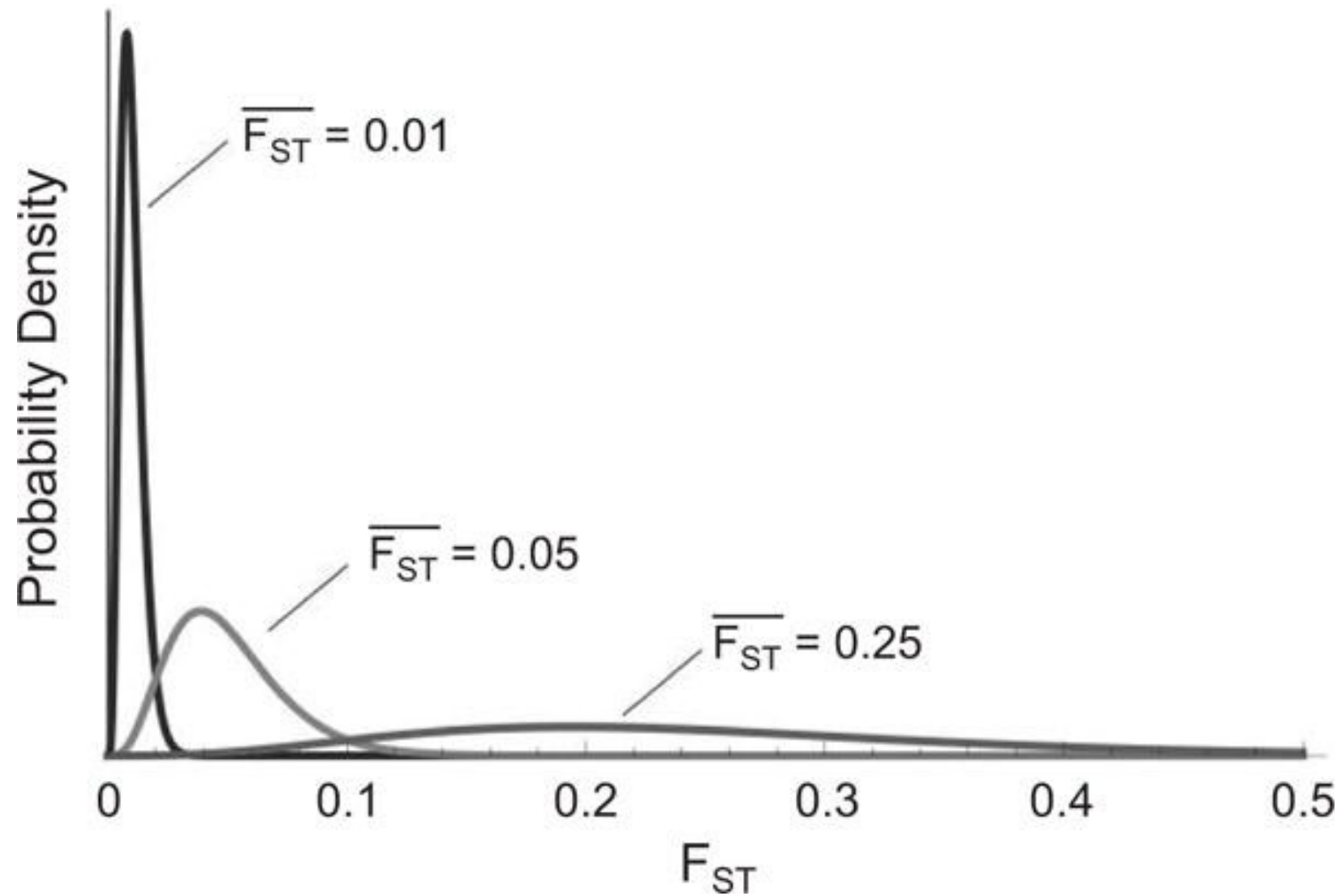
Schematic of *EPAS1* function



Tibetans live in the World's highest mountains



People sometimes use Tajima's D as a summary statistic of the SFS.

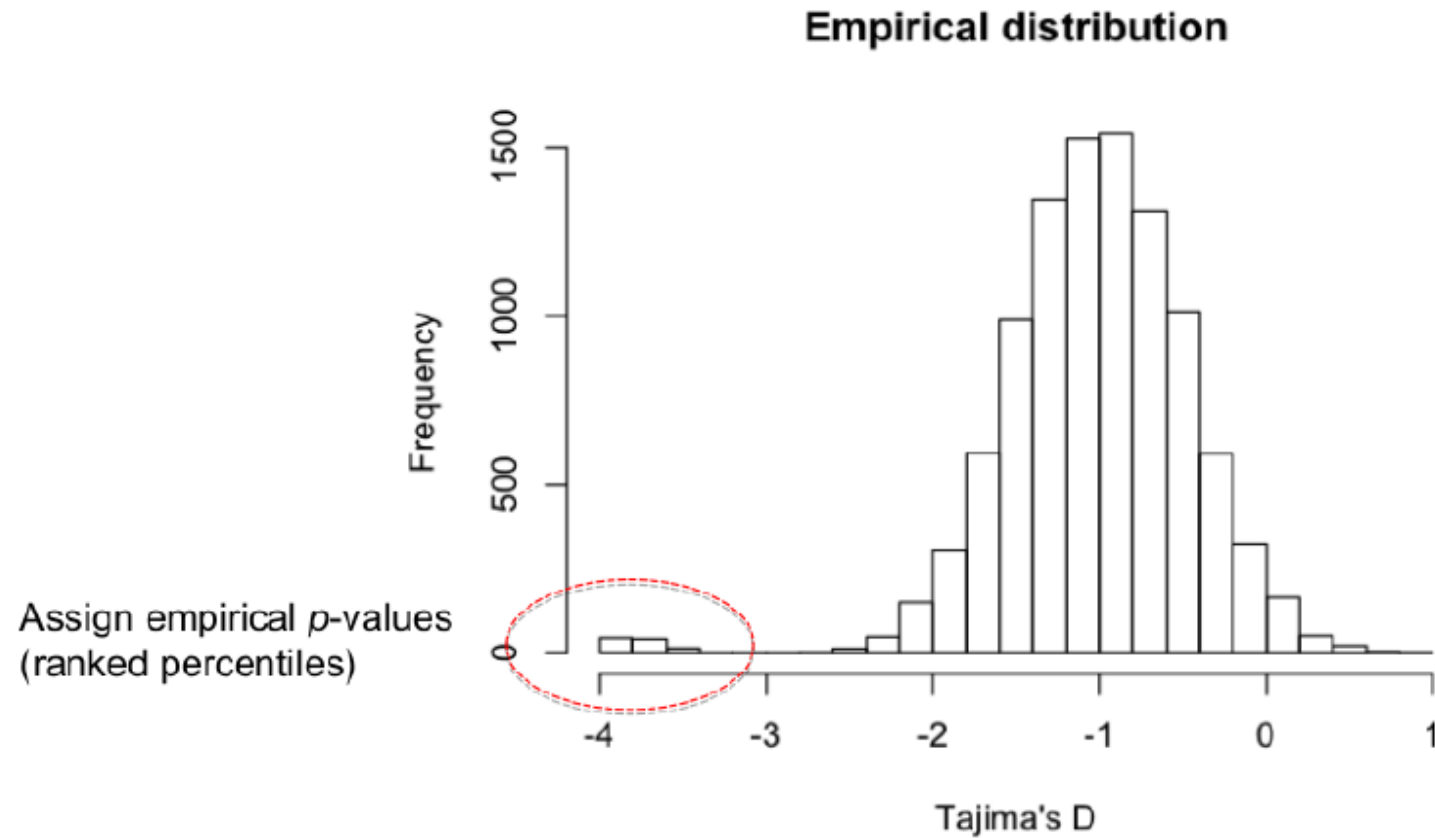


Even under neutrality, genome-wide F_{ST} varies

Assessing significance

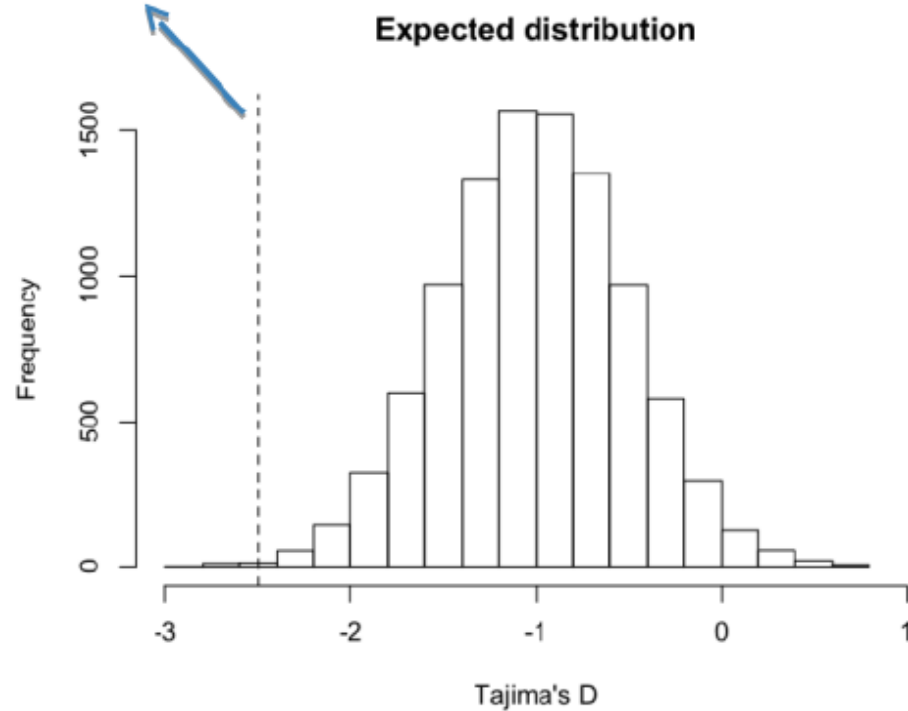
1. Calculate empirical p-values and False Discovery Rate.
2. Model demographic history to create null distribution.
3. Evaluate genomic context, e.g. local recombination rate.
4. Cross-validate with complementary scan statistics.

1. Empirical p-value

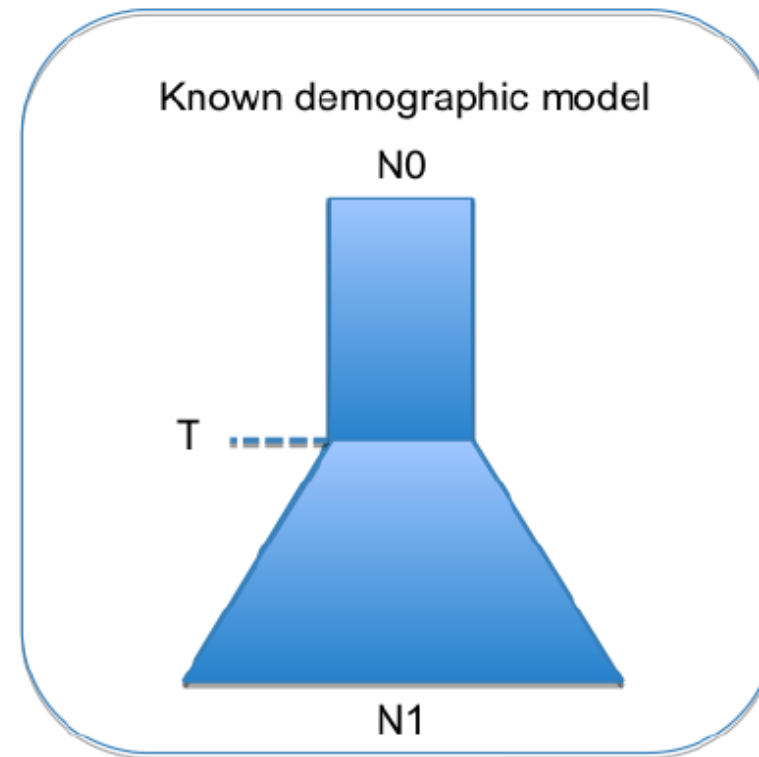


2. Model neutral distribution

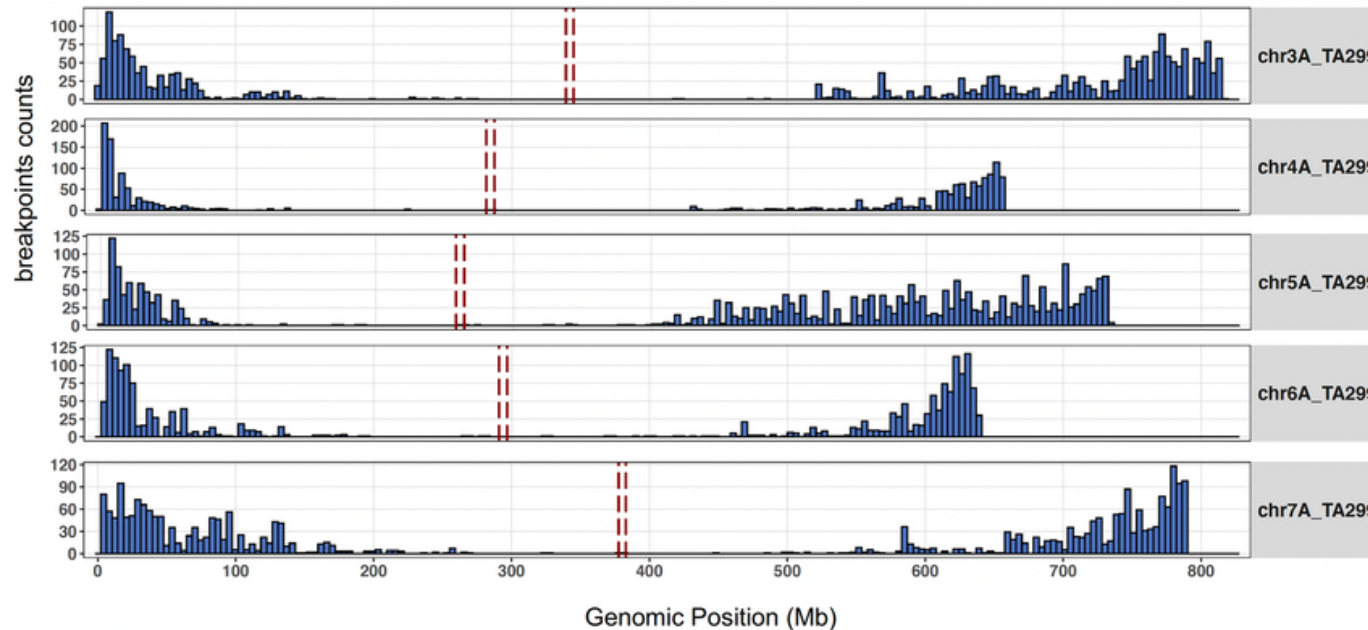
Assign p -values
(based on ranked percentile of observed value)



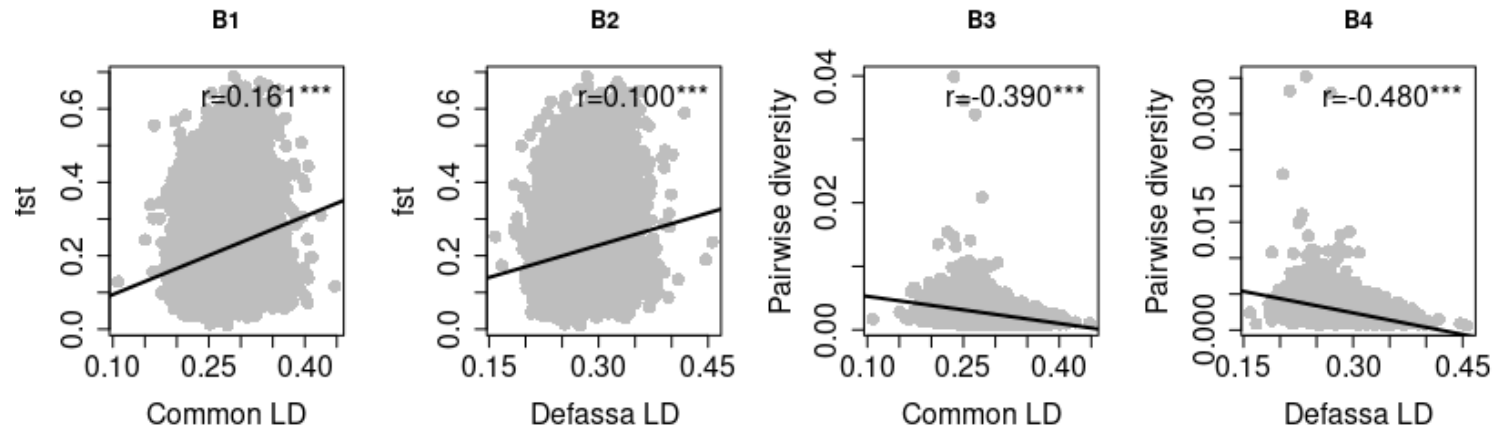
Genome / Chromosome



3. Evaluate genomic context



Recombination rates vary over chromosomes.



Regions of low recombination often look like regions under positive selection.

4. Use complementary scan statistics

